

COMPLETE FEATURE LIST

Eko Excel OS & SmartSchool

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1. EKO EXCEL OS - COMPLETE FEATURE SPECIFICATION

1.1 STUDENT LEARNING FEATURES

- **Academic Heatmap Dashboard:** Real-time mastery levels by topic. RED (<40%), YELLOW (40-70%), GREEN ($\geq 70\%$). Updates after every quiz.
- **Brain Map Profiler:** Detects learning style (Visual/Auditory/Kinesthetic), peak cognitive hours, optimal sprint duration.
- **WAEC Readiness Meter:** Live WAEC score prediction. Shows projected final score and gap to passing.
- **Fix Pack Engine:** Generates 6-minute personalized lessons: 60-sec refresher + 3-min lesson + 2-min challenge.
- **Offline Mastery Companion:** Complete functionality without internet. Syncs when connected.
- **Localized Exam Simulator:** WASSCE/WAEC mock tests with exact exam format. Offline functionality.
- **Adaptive Difficulty Adjuster:** Real-time difficulty adjustment based on performance. Prevents frustration.
- **Root Cause Diagnosis Engine:** Identifies prerequisite failures. E.g., Quadratic weakness caused by Variables weakness.
- **Personalized Remediation Pathway:** Custom practice tracks for students with learning gaps.
- **Gamified Progress Tracking:** Celebrations, badges, achievements, leaderboards. Motivational notifications.
- **Content Modality Matching:** Delivers content in optimal format: visual, auditory, or kinesthetic.
- **Micro-Assessment Battery:** Diagnostic quizzes to pinpoint exact misconceptions (not graded).
- **Concept Prerequisite Mapping:** Internal curriculum graph showing concept dependencies.
- **Performance History Tracking:** Stores all quiz attempts with detailed analytics.
- **Notification Scheduling Engine:** Sends notifications at optimal times based on learning pattern.

1.2 TEACHER EFFICIENCY FEATURES

- **Zero-Marking AI Engine:** 3-click: Select Topic → Set Difficulty → Enter Count. Auto-generates, grades, provides feedback.
- **Automatic Question Generation:** Standardized questions at any difficulty level from verified question bank.

- **Automatic Grading & Feedback:** AI grades all assignments and provides diagnostic feedback instantly.
- **Classroom Diagnostic Dashboard:** Real-time view of struggling students. Color-coded by mastery level.
- **Real-Time Comprehension Metrics:** Post-assessment: class average, topic mastery, students needing help.
- **Skill-Gap Flagging System:** Automatic alerts when students fall below mastery thresholds.
- **Automated Remediation Recommender:** Suggests Fix Packs or exercises for struggling cohorts.
- **Digital Lesson Sync:** Standardized lesson scripts on teacher devices. Ensures consistent teaching.
- **Attendance Tracking System:** QR code check-in. Automatic attendance and presence logging.
- **Performance Analytics by Student:** Identifies improving, plateauing, or declining students. Early at-risk detection.
- **Class-Level Progress Tracking:** Shows class mastery of topics. Helps plan reteaching.
- **Time-on-Task Monitoring:** Tracks student engagement time. Flags disengagement.
- **Fix Pack Assignment Workflow:** One-click assign, track completion, send reminders.
- **Teacher Task Scheduler:** Automated reminders for grading, planning, assessments.
- **Parent Communication Hub:** Central place to message parents via SMS/WhatsApp.

1.3 PARENT ENGAGEMENT FEATURES

- **WhatsApp Weekly Reports:** Automated SMS/WhatsApp every Sunday with progress, weak topics, WAEC projection.
- **Academic Heatmap Sharing:** Parents see child's heatmap. Red/Yellow/Green visualization. Easy to understand.
- **WAEC Readiness Prediction:** Shows projected score, what needs improvement, timeline to passing.
- **Real-Time Performance Alerts:** Instant notification when child falls behind or excels.
- **Remediation Action Plans:** Clear, actionable steps parents can take at home.
- **Real-Time Attendance Alerts:** SMS/WhatsApp for check-in/out. Prevents skipping.
- **Digital Report Cards:** Instant access to terminal reports. No paper delays.
- **Parent Progress Dashboard:** Secure login to view all child data: quizzes, grades, trends, projections.
- **Two-Way Messaging:** Parents message teacher/school with concerns.
- **Milestone Celebrations:** Notifications when child achieves topic mastery.
- **Comparative Analytics (Opt-in):** Anonymized comparison against class/school/state averages.
- **WAEC Eligibility Alerts:** Notifies of minimum attendance requirements for WAEC eligibility.

1.4 SCHOOL ADMINISTRATION FEATURES

- **School-Level Compliance Hub:** Principal dashboard: teacher attendance, lesson completion, assessment data, WAEC readiness.
- **Resource Allocation Monitor:** Tracks government-supplied materials. Flags shortages.
- **Academic Performance Tracking:** WAEC readiness by class, subject, student. Identifies underperformers.
- **Teacher Performance Analytics:** Aggregates student improvement data for fair evaluation.
- **Enrollment & Retention Analytics:** Tracks churn risk, enrollment growth, capacity.
- **Asset & Inventory Tracker:** Manages infrastructure inventory with location tracking.
- **Staff Performance Monitoring:** Tracks attendance, lesson completion, student outcome improvements.
- **Curriculum Coverage Auditor:** Shows which topics taught by which teacher on which dates.
- **Automated Report Generation:** One-click termly reports: academic, staff, resources, WAEC projections.
- **Quality Assurance Alerts:** Flags anomalies: score drops, unusual patterns, incomplete assessments.

1.5 DISTRICT SUPERVISION FEATURES

- **Zonal Performance Heatmaps:** Geographic visualization by zone/LGA. Identifies underperforming regions.
- **School Comparison Analytics:** Compare schools against district/state averages.
- **Supervisor Audit Log:** Digital record of inspection visits, QA assessments, findings.
- **Teacher Quality Dashboard:** District-wide teacher effectiveness aggregated.
- **Resource Equity Monitor:** Identifies resource gaps. Supports allocation decisions.

- **District WAEC Forecasting:** Predicts results 3-6 months ahead. Identifies schools needing help.
- **Curriculum Compliance Checker:** Verifies curriculum teaching, tracks pacing.
- **Intervention Recommendation Engine:** AI suggests which schools need targeted support.
- **Professional Development Tracking:** Tracks training completion, certification status.
- **District Performance Reports:** Monthly/quarterly reports on trends, resources, teaching quality.

1.6 MINISTRY GOVERNANCE FEATURES

- **State-Wide Governance Dashboard:** Macro view of education across all schools. WAEC readiness by state/zone/district.
- **Macro Revenue & Budget Protection:** Tracks fund allocations. Monitors PPP efficiency. Identifies leakages.
- **Predictive Human Capital Forecasting:** Forecasts state workforce readiness. Informs policy.
- **Statewide Benchmarking:** Shows top-performing districts. Facilitates peer learning.
- **Teacher Shortage Identification:** Aggregates teacher ratios. Identifies schools needing teachers.
- **Infrastructure Gap Analysis:** Comprehensive view of physical infrastructure across all schools.
- **Policy Impact Evaluation:** Tracks policy outcomes. A/B tests interventions.
- **Equity & Access Monitor:** Identifies disparities by region/ethnicity/gender.
- **Automated Compliance Auditing:** Continuous monitoring with automated alerts.
- **Strategic Planning Intelligence:** Key insights for 3-year and 5-year planning.
- **Public Accountability Reports:** Public-facing reports. Increases transparency.
- **QR Code Attendance System:** Mandatory QR check-in for teachers. Prevents proxy signatures.

2. SMARTSCHOOL - COMPLETE FEATURE SPECIFICATION

2.1 TEACHER INFRASTRUCTURE

- **AI Homework Generator:** Generates customized homework based on classroom gaps. Aligned to curriculum.
- **Automated Marking Engine:** Scans, scores assessments instantly. Supports MCQ, short answer, essays.
- **Teacher Diary & Performance Log:** Tracks instructional pacing, lesson dates, engagement metrics.
- **Micro-Concept Lesson Planners:** Modular, curriculum-aligned guides optimized for high-density classrooms.
- **Diagnostic Dashboard:** Real-time classroom comprehension metrics post-assessment.
- **Skill-Gap Flagging System:** Alerts when students fall behind on prerequisites.
- **Automated Remediation Recommender:** Suggests targeted exercises for struggling cohorts.
- **Lesson Video Library:** Curriculum-aligned videos. Can be assigned or shown in class.
- **Interactive Whiteboard Integration:** Push content to smart boards. Real-time polling.
- **Assessment Item Bank:** 10,000+ verified questions searchable by topic, difficulty, type.
- **Attendance & Participation Tracker:** Digital attendance. Tracks class participation, group work.
- **Class Management Tools:** Roster management, seating, group assignments. Auto-generate name cards.
- **Grade Book Management:** Automated grade calculation with weighted averages.
- **Communication Hub:** Send messages to students/parents/admin via SMS/WhatsApp.
- **Professional Development Log:** Tracks training, certifications earned.

2.2 STUDENT LEARNING INFRASTRUCTURE

- **Offline Cognitive Engine:** Complete functionality without internet. Syncs when connected.
- **The Brain Map:** Personalized profile tracking strengths and misconceptions.
- **WAEC Readiness Meter:** Predictive machine forecasting exam passing probability.
- **Automated Mastery Adjuster:** Dynamically alters content speed and difficulty.
- **AI-Powered Diagnostic Testing:** Micro-assessments pinpointing root causes.
- **Gamified Adaptive Quizzes:** Modular practice with progress bars, badges, achievements.
- **Bite-Sized Conceptual Videos:** Low-bandwidth 3-5 minute explainers mapped to curriculum.
- **Interactive Simulations:** Virtual labs for science, interactive models for math.
- **Peer Collaboration Tools:** Forum for questions, student help, teacher moderation.
- **Study Group Features:** Students form groups, track progress, share notes.
- **Progress Visualization:** Student-friendly dashboards with charts, milestones.
- **Personalized Recommendations:** System recommends next topics based on mastery.
- **Offline Content Sync:** Auto-syncs new content when internet available.
- **Performance History:** Students view performance trends. Motivates focused effort.

2.3 PARENT ENGAGEMENT INFRASTRUCTURE

- **Automated WhatsApp Engagement:** Instant updates on performance, attendance, milestones.
- **Academic Heatmaps:** Visual dashboards mapping subject mastery and learning decay.
- **Exam Readiness Notifications:** Alerts showing projected score and gap to passing.
- **Remediation Action Plans:** Clear steps parents can support at home.
- **Real-Time Attendance Alerts:** SMS/WhatsApp for check-in/out. Prevents skipping.
- **Digital Report Cards:** Instant access to terminal results. Downloadable PDF.
- **Parent Login Portal:** Secure access to view all child data.
- **Performance Comparison:** Anonymized comparison against class/school averages.
- **Two-Way Messaging:** Parents message teacher/admin with concerns.
- **Fee Payment Tracking:** Shows fees due, payment history, pending balances.
- **School Event Notifications:** Alerts about events, holidays, exam dates, meetings.
- **Child Safety Alerts:** Notifications about disciplinary actions, health concerns.
- **Learning Progress Videos:** Optional weekly videos showing child's progress.

2.4 SCHOOL MANAGEMENT INFRASTRUCTURE

- **The Result Gate:** Revenue protection: report access restricted until fees cleared.
- **Automated FinOps Dashboard:** Manages payroll, invoices, fee collections.
- **Academic Audit Trail:** Monitors teacher lesson completion, performance, coverage.
- **Enrollment & Retention Analytics:** Tracks churn risk, growth, capacity.
- **Staff Performance Analytics:** Aggregates student data for teacher evaluation.
- **Asset & Inventory Tracker:** Manages school infrastructure and equipment.
- **Fee Management System:** Tracks fees, generates invoices, identifies defaulters.
- **Payroll Management:** Automated monthly payroll, salary slips, deductions.
- **School Calendar Management:** Central calendar synced to all users.
- **Timetable & Scheduling:** Optimized timetables, room usage tracking.
- **Complaint & Grievance System:** Digital system for recording and resolving complaints.
- **Document Management:** Central repository for school documents. Version-controlled.
- **Visitor Management:** Tracks visitor sign-in/out, purpose, duration.
- **Maintenance Request System:** Staff submit requests, tracked to completion.

3. CROSS-PRODUCT FEATURES

- **Multi-Language Support:** English, Yoruba, Igbo, Hausa interface support.
- **Offline Functionality:** Complete offline operation with sync when connected.
- **Mobile-First Design:** Optimized for phones. Works on WhatsApp, USSD, SMS.
- **Accessibility Features:** High contrast, text size adjustment, screen reader compatible.
- **Data Security & Privacy:** End-to-end encryption. GDPR compliant. Never share without consent.
- **Role-Based Access Control:** Users see only relevant data for their role.
- **Automated Backup:** Daily backups with disaster recovery.
- **API Integration:** RESTful APIs for integration with other systems.
- **Analytics & Reporting:** Real-time dashboards, scheduled reports, custom builder.
- **System Health Monitoring:** Uptime monitoring, performance metrics, error logging.
- **Continuous Updates:** Automatic bug fixes and improvements. Zero downtime.
- **Training & Support:** Video tutorials, guides, 24/7 helpline, email support.

4. TECHNICAL SPECIFICATIONS

- **Technology Stack:** Frontend: React 18 + TypeScript | Backend: NestJS | Database: PostgreSQL | Cache: Redis | Queue: Bull | Real-time: Socket.io
- **Scalability:** 1M+ concurrent students. Auto-scaling. Load-balanced. CDN delivery.
- **Performance:** Page load <2sec. Quiz submit <100ms. Notifications <5sec. 99.9% uptime.
- **Mobile Compatibility:** iOS 12+, Android 8+. App size <50MB. Works on 2G. Min 512MB RAM.
- **Data Retention:** Student records: indefinite. Logs: 7 years. System logs: 90 days.
- **API Rate Limits:** 10,000 requests/hour per school with throttling after limit.
- **Database Design:** Normalized schema with indexing. Optimized queries for analytics.
- **Infrastructure:** Cloud-hosted (AWS/Google). Auto-scaling. Multi-region redundancy.

5. DATA REQUIREMENTS

- **Student Data:** Full name, age, gender, admission number, class, parent contact, medical info, emergency contact, SEN status.
- **Teacher Data:** Full name, employee ID, qualification, specialization, experience, contact, employment status, payroll.
- **Assessment Data:** Quiz ID, questions, responses, score, time, submission timestamp, teacher ID.
- **Content Data:** Curriculum topics, learning objectives, difficulty ratings, 10,000+ question items.
- **School Data:** Name, address, registration, principal, contact, zone, district, capacity.
- **Parent Data:** Full name, occupation, contact, relationship, emergency contact, fee status.
- **Historical Data:** Previous WAEC results, enrollment patterns, teacher performance, payment history.

---END OF FEATURE SPECIFICATION---